DAW - WEB INTERFACE DESIGN

UNIT 3.
USE OF STYLES

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CSS STYLE SHEETS

Objectives:

- Analyze the different options for presenting information in web documents.
- Assess the importance of defining and applying a style guide in web application development.
- Select colors and other design elements for on-screen display.
- Create and use design templates.
- Recognize the possibility of modifying HTML tags.
- Define styles directly.
- Apply global styles using external style sheets.
- Identify the different properties of each element.
- Create style classes.

CSS STYLE SHEETS (II)

4.1. Introduction to CSS
4.2. How to Use External Style Sheets
4.3. Media: Adapting to the Viewing Environment or Device
    4.3.1. @media
    4.3.2. @import
4.4. The CSS Box Model
4.5. Inheritable CSS Properties
4.6. CSS3
4.7. Bootstrap 4: Responsive and Attractive Design
    4.7.1. Containers
    4.7.2. Grid System: Columns in Bootstrap
    4.7.3. Typography
    4.7.4. Colors
4.8. YUI (Yahoo! User Interface)

INTRODUCTION TO CSS

When working with web pages written in HTML and XHTML, it is desirable for them to have a specific style and a well-defined appearance.

The most efficient way to achieve this is to separate the part where the content is placed, represented by files programmed in HTML, from the part of the project that contains aspects related to presentation.

This is where CSS (Cascading Style Sheets) comes in, a style sheet language responsible for controlling the appearance of websites.

The use of CSS style sheets provides numerous advantages: it improves the accessibility of the document, reduces the complexity of maintenance, whether it is carried out by the original developer or by third parties, and also facilitates the display of the website on different devices.

BROWSER ENGINES AND CSS SUPPORT

- Not all browsers implement CSS features at the same pace.
- The rendering engine is the component responsible for interpreting HTML, CSS, and JavaScript and displaying web content.
- Modern browsers offer a high level of compatibility with current CSS standards.
- Main Rendering Engines
  - Blink → Chrome, Edge, Opera
  - Gecko → Firefox
  - WebKit → Safari

INTERNAL CSS STYLES

- CSS styles can be added directly to an HTML page using the <style> tag.
- The <style> section is typically placed inside the <head> element.
- This approach is useful for learning or styling a single page.
- For larger websites, it is recommended to use external CSS files.

```
<html>
 <head>
  <meta http-equiv="Content-Type" content="text/html; charset=iso-8859
  -1"/>
  <title>
    Ejemplo de estilos CSS en el propio documento
  </title>
  <style type="text/css">
    p {
      color: black; font-family: Verdana;
    }
  </style>
 </head>
 <body>
  <p>Un parrafo de texto.</p>
 </body>
</html>
```

INTERNAL CSS STYLES (II)

FUNDAMENTAL

The main disadvantage of not using CSS files and instead defining styles directly within the HTML code is that, if a change to the defined styles is required, all pages that include the style to be modified must be updated.

HOW TO USE EXTERNAL STYLE SHEETS

Nowadays, external style sheets are commonly used.

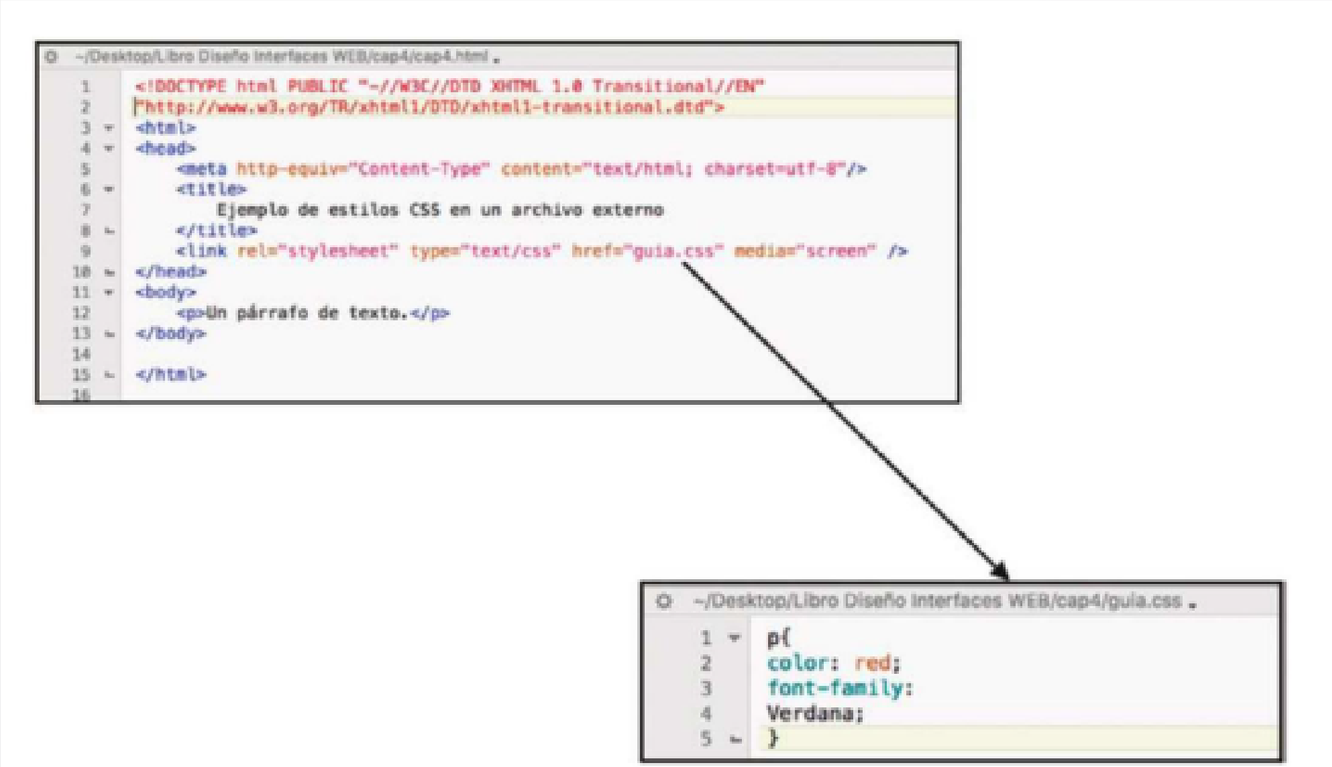
In this case, instead of having the code responsible for styling the website within the HTML document itself, that code is stored in a separate file.

From the main HTML page, the CSS file is accessed through the <link> tag.

The only particular aspect to keep in mind is that the file containing the styles must have the .css extension.

As many CSS files as needed can be linked to a web page

HOW TO USE EXTERNAL STYLE SHEETS (II)



HOW TO USE EXTERNAL STYLE SHEETS (III)

The purpose of each attribute used to link an HTML document to a CSS style sheet is as follows:

- **rel:** Specifies the type of relationship between the linked resource (the CSS file) and the HTML document. For CSS files, the value is stylesheet.
- **type:** Specifies the type of the linked resource. For CSS files, the value is text/css.
- **href:** Specifies the URL of the CSS file. This URL can be either relative or absolute and may point to an internal or external resource.
- **media:** Specifies the media type to which the CSS styles will be applied. CSS media types and their operation will be explained in more detail later.

HOW TO USE EXTERNAL STYLE SHEETS (IV)

Nowadays, the use of CSS in HTML is widespread because a single CSS file can be used to apply the same style guide to all the HTML pages of a website.

In this way, one of the most important principles of graphic design is achieved: **maintaining a consistent style across all the pages that make up the site**.

In addition, if a common style change is required across all pages, it is sufficient to modify the style sheet, and the changes will automatically be applied throughout the entire website linked to that CSS file.

HOW TO USE EXTERNAL STYLE SHEETS (V)

- Ejemplo:

```
<html>
 <head>
  <meta http-equiv="Content-Type" content="text/html; charset=UTF-8"/>
    <title>
            Ejemplo de estilos CSS en el propio documento
    </title>
    <link rel="stylesheet" type="text/css" media="screen, handheld"
    href="hojaEst.css">
 </head>
 <body>
  <p>Un párrafo de texto.</p>
 </body>
</html>
```

El código CSS (hojaEst.css):

```
p {
        color: red; font-family: Verdana;
}
```

MEDIA: ADAPTING TO THE VIEWING MEDIUM OR DEVICE

Style sheets make it possible t**o specify how a document is displayed depending on the type of viewing medium**.

These media refer to the channels through which a website is presented, such as audio, visual displays, printed documents, and others.

It is common practice to define a specific property for each type of medium.

To specify the target medium, several mechanisms can be used, either within the CSS style sheet or directly in the website's HTML file.

MEDIA: ADAPTING TO THE VIEWING MEDIUM OR DEVICE (II)

Media Type	Description	Current Status
all	All media types.	✅ Commonly used (default)
print	Printers and print preview.	✅ Commonly used
braille	Tactile devices using the Braille system.	❌ Deprecated
embossed	Braille printers.	❌ Deprecated
handheld	Handheld devices such as mobile phones and PDAs.	❌ Deprecated
projection	Projectors and presentation devices.	❌ Deprecated
screen	Computer, tablet, and smartphone screens.	✅ Most commonly used
speech	Speech synthesizers and screen readers.	✅ Supported
tv	Televisions and low-resolution display devices.	❌ Deprecated

@MEDIA

The @media rule specifies the target media types. Through this feature, a single style sheet can contain rules for different media. If the same style applies to multiple media, their names can be specified as a comma-separated list.

In modern web development, media queries are mostly used to create **responsive designs** based on screen characteristics rather than media types:

```
/* Mobile */
@media (max-width: 768px) {
    .menu {
        display: none;
    }
}
```

```
/* Dark mode */
@media (prefers-color-scheme: dark) {
    body {
        background: #222;
        color: white;
    }
}
```

```
/* Printing */
@media print {
    nav,
    footer {
        display: none;
    }
}
```

@IMPORT

Using @import, it is possible to import style rules from other CSS files and specify the media or media types to which the imported styles should apply. In this case, unlike @media, the URL of the CSS file containing the style sheet is specified first, followed by the target medium to which the styles will be applied. It is important to note that @import rules must appear before all other style rules.

If the styles from the imported file are to be applied to multiple media types simultaneously, the names of those media types should be separated by commas.

```
@import url("filename.css") mediaType;
```

```
@import "ficheroEstilo1.css" print;
@import "ficheroEstilo2.css" screen;
@import "ficheroEstilo3.css" print, screen;
@import "ficheroEstilo4.css"; (Por defecto se aplica a todo)
```

@IMPORT (II)

Although the @import rule is still supported by current browsers, external style sheets are typically linked using the <link> element in the HTML document. The @import rule is mainly used in specific cases, such as organizing CSS files within a larger stylesheet.

THE CSS BOX MODEL

The box model is one of the most important features of style sheets.

When an HTML page is created, the use of each tag simulates what could be considered a box, giving each one an identifier indicated by the tag name. This box system makes it possible, from the CSS file, to select each box and apply the desired style to it.

Using CSS style sheets allows the modification of all the characteristics of each of the boxes that have been created. The following section analyzes each of these parts through an example that incorporates new properties

THE CSS BOX MODEL (II)

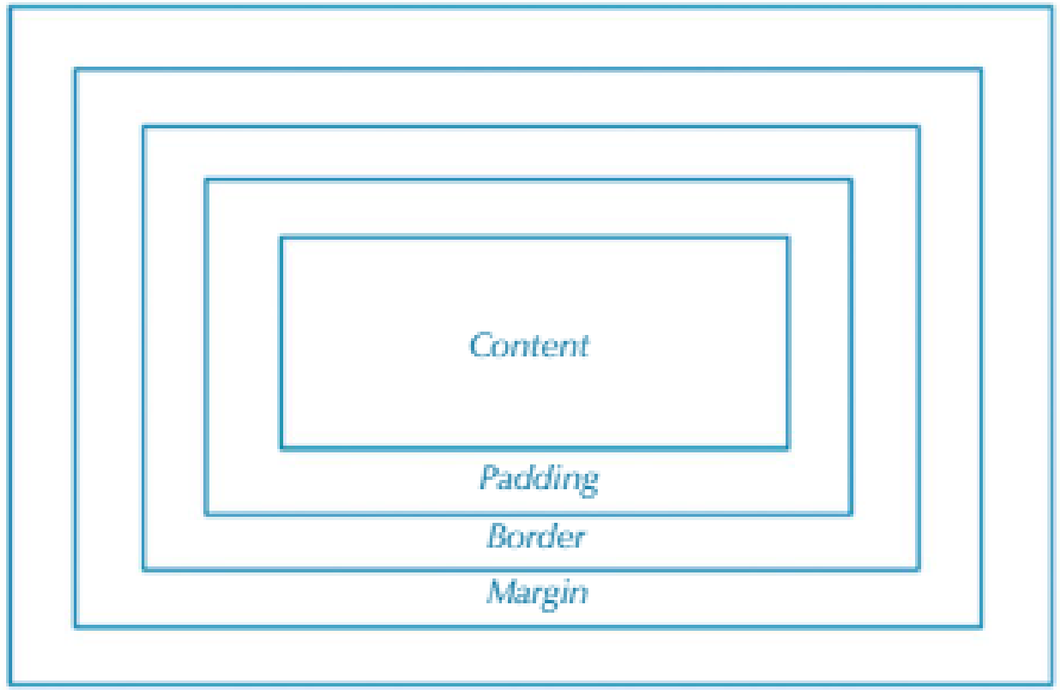


Figura 4.3
Diagrama modelo de cajas en CSS.

¿ANY QUESTION?